MA1522 Linear Algebra for Computing

Lesson 15: QR Factorization and Least Squares Solutions

$$-I_3 = \begin{pmatrix} -1 & & \\ & -1 & \\ & & -1 \end{pmatrix}.$$

QR decomp:

$$Q = \begin{pmatrix} -1 & & \\ & -1 & \\ & & -1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} = R.$$

$$\begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} -1 & & \\ & -1 & \\ & & -1 \end{pmatrix}$$

Consider functions on $[-1, 1]$.
(continuous, real valued)

We can define inner product:

$$\langle f, g \rangle := \int_{-1}^{1} f(x) g(x)\, dx. \quad \left(\langle f, f \rangle = 0 \Leftrightarrow f = 0 \right)$$

In this sense, if $f(x)$ is even ($f(-x) = f(x)$)
$g(x)$ odd ($g(-x) = -g(x)$).

then $f(x)g(x)$ is odd, so $\int_{-1}^{1} f(x)g(x)dx = 0$.

Announcements

- No class on Friday 9 October (Well-Being Day), nor on Tuesday 13 and Friday 16 October. The next lesson is **Tuesday 20 October**.

- **Online Quiz 4** opens **Saturday 17 October, 0:00**, and is due **Sunday 18 October, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson15` (e.g. `A1234567Z_lesson15.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 15**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev

$$A = \begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_k \\ | & | & & | \end{pmatrix}$$

$$A^T A = \begin{pmatrix} & u_i^T \cdot u_j & \end{pmatrix}$$

$$= \begin{pmatrix} & u_i \cdot u_j & \end{pmatrix}$$



So A is orthogonal $\Leftrightarrow A^T A = I \Leftrightarrow \begin{cases} u_i \cdot u_j = 0 & \text{if } i \neq j \\ u_i \cdot u_i = 1 \end{cases}$.

$\{u_1, \ldots, u_k\}$ is orthogonal.

orthonormal.

Question 1

(QR factorization is reformulation of Gram-Schmidt process).

$$A = \begin{pmatrix} 1 & -3 & -1 \\ 0 & 2 & 4 \\ 2 & -2 & -3 \\ 2 & -1 & -1 \end{pmatrix} = (\,\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\,).$$

The Gram-Schmidt process applied to $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ gives the orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for $\mathrm{Col}(A)$, where

$$\mathbf{v}_1 = \mathbf{a}_1, \quad \mathbf{v}_2 = \mathbf{a}_2 + \mathbf{v}_1, \quad \mathbf{v}_3 = \mathbf{a}_3 + \mathbf{v}_1 - \mathbf{v}_2.$$

$= \mathbf{a}_2 - \frac{\mathbf{a}_2 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2}\mathbf{v}_1$ (second step in Gram-Schmidt)

3×3

1. Find two upper triangular matrices R_1, R_2 such that (first step in Gram-Schmidt)

$$(\,\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\,)\,R_1 = (\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{a}_3\,), \qquad (\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{a}_3\,)\,R_2 = (\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3\,).$$

2. Deduce that $A = QR$, where Q is ~~orthogonal~~ and R is upper triangular with positive diagonal entries.

$Q^TQ = I_3$..

Question 1 — solution: part 1

$$\begin{cases} v_1 = a_1 \\ v_2 = a_2 + v_1 \\ v_3 = a_3 + v_1 - v_2 \end{cases}$$

$v_1 = a_1$

$$(a_1 \; a_2 \; a_3) R_1 = (v_1 \; v_2 \; a_3)$$

$$(v_1 \; v_2 \; a_3) R_2 = (v_1 \; v_2 \; v_3)$$

$$(v_1 \; v_2 \; a_3) = (v_1 \quad a_2 + 1 v_1 \quad a_3) = (v_1 \; a_2 \; a_3) \begin{pmatrix} 1 & 1 & \\ & 1 & \\ & & 1 \end{pmatrix} \qquad R_1 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(v_1 \; v_2 \; v_3) = (v_1 \quad v_2 \quad a_3 + 1 v_1 + (-1) v_2)$$

$$= (v_1 \; v_2 \; a_3) \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \qquad R_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

We modify a_3, so these coefficients are in the third col.

Question 1 — solution: part 2

2. Deduce that $A = QR$, where Q is orthogonal and R is upper triangular with positive diagonal entries.

From part 1: $A = \begin{pmatrix} | & | & | \\ a_1 & a_2 & a_3 \\ | & | & | \end{pmatrix}$

$$AR_1R_2 = \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix} = Q' \quad (1)$$

Question: Is $Q' = \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix}$ orthogonal? No, because $\{v_1, v_2, v_3\}$ is orthogonal, but NOT orthonormal.

A matrix $A = (u_1 \cdots u_k)$ is orthogonal ($A^TA = I$) if and only if $\{u_1, \ldots, u_k\}$ is orthonormal.

So $Q = \left(\frac{v_1}{\|v_1\|} \quad \frac{v_2}{\|v_2\|} \quad \frac{v_3}{\|v_3\|}\right)$ is "orthogonal". $Q^TQ = I_3$.

$= Q' \begin{pmatrix} \frac{1}{\|v_1\|} & & \\ & \frac{1}{\|v_2\|} & \\ & & \frac{1}{\|v_3\|} \end{pmatrix}$ (3) $\Leftrightarrow Q \begin{pmatrix} \|v_1\| & & \\ & \|v_2\| & \\ & & \|v_3\| \end{pmatrix} = Q'$.

$AR_1R_2 \overset{(1)}{=} Q' \overset{(2)}{=} Q \begin{pmatrix} \|v_1\| & & \\ & \|v_2\| & \\ & & \|v_3\| \end{pmatrix} \Longrightarrow A = Q \begin{pmatrix} \|v_1\| & & \\ & \|v_2\| & \\ & & \|v_3\| \end{pmatrix} R_2^{-1}R_1^{-1}$.

$= R = \begin{pmatrix} \|v_1\| & * & * \\ 0 & \|v_2\| & * \\ 0 & 0 & \|v_3\| \end{pmatrix}$

Section 5.4 QR Factorization

Theorem (QR Factorization)

Suppose A is an $m \times n$ matrix with linearly independent columns. Then A can be written as

$$A = QR$$

for some $m \times n$ matrix Q such that $Q^T Q = I_n$ and invertible upper triangular matrix R with positive diagonal entries. This decomposition is called a **QR factorization** of A.

Algorithm to QR Factorization

Let $A = (\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n)$ have linearly independent columns.

1. Perform Gram-Schmidt on $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$ to obtain an orthonormal set $\{\mathbf{q}_1, \ldots, \mathbf{q}_n\}$.
2. Set $Q = (\mathbf{q}_1 \ \mathbf{q}_2 \ \cdots \ \mathbf{q}_n)$.
3. Compute $R = Q^T A$.

$A = QR \implies Q^T A = (Q^T Q)R = R$.

Exercise: Write Q and R explicitly for Question 1.

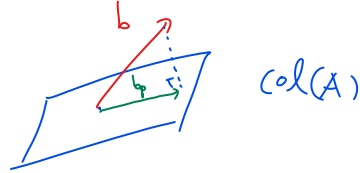
Challenge 1

Let A be an $m \times n$ matrix with linearly independent columns. Show that the QR factorization of A is unique: if $A = QR = Q'R'$, where $Q^T Q = Q'^T Q' = I_n$ and R, R' are upper triangular with positive diagonal entries, then $Q = Q'$ and $R = R'$.

Section 5.5 Least Squares Approximation

Recall $A\mathbf{x} = \mathbf{b}$ is consistent $\iff \mathbf{b} \in \text{col}(A)$.

$A\mathbf{x} = \mathbf{b}_p$



Let A be an $m \times n$ matrix and $\mathbf{b}$ a vector in $\mathbb{R}^m$.

Definition

A vector $\mathbf{u}$ in $\mathbb{R}^n$ is a **least squares solution** of $A\mathbf{x} = \mathbf{b}$ if for every vector $\mathbf{v} \in \mathbb{R}^n$,

$$\|A\mathbf{u} - \mathbf{b}\| \leq \|A\mathbf{v} - \mathbf{b}\|.$$

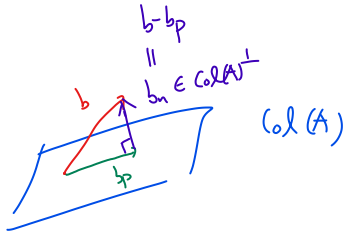
Theorem

$A\mathbf{u} = \mathbf{b}_p$.

A vector $\mathbf{u}$ in $\mathbb{R}^n$ is a least squares solution to $A\mathbf{x} = \mathbf{b}$ if and only if $A\mathbf{u}$ is the projection of $\mathbf{b}$ onto $\text{Col}(A)$.

Section 5.5 Least Squares Solutions and $A^TA\mathbf{x} = A^T\mathbf{b}$

Let A be an $m \times n$ matrix and $\mathbf{b}$ a vector in $\mathbb{R}^m$.



Theorem

A vector $\mathbf{u}$ in $\mathbb{R}^n$ is a least squares solution to $A\mathbf{x} = \mathbf{b}$ if and only if $\mathbf{u}$ is a solution to $A^TA\mathbf{x} = A^T\mathbf{b}$.

pf.

u is a least squares sol $\iff$ Au is the orthogonal projection of b on $Col(A)$.

$\iff$ $b - Au \in Col(A)^\perp$.

$\iff$ $A^T(b - Au) = 0$ $\iff$ $A^TAu = A^Tb$ $\square$.

Question 2

Let

$$B = \begin{pmatrix} -1 & 1 & 0 \\ 2 & 1 & 3 \\ -2 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} -4 \\ 1 \\ -3 \\ -3 \end{pmatrix}.$$

1. Find all least squares solutions of $B\mathbf{x} = \mathbf{b}$.

2. Find the projection of $\mathbf{b}$ onto $\mathrm{Col}(B)$. Is $B\mathbf{x} = \mathbf{b}$ consistent?

Question 2 — solution: part 1

$B = \begin{pmatrix} -1 & 1 & 0 \\ 2 & 1 & 3 \\ -2 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix}$. $b = \begin{pmatrix} -4 \\ 1 \\ -3 \\ -3 \end{pmatrix}$. By thm, it B equiv to solve $B^T Bx = B^T b$.

$$\left(B^T B \mid B^T b \right) = \left(\begin{array}{ccc|c} 10 & -5 & 5 & 15 \\ -5 & 10 & 5 & -15 \\ 5 & 5 & 10 & 0 \end{array} \right) \xrightarrow{\text{rref}} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Solutions to $B^T Bx = B^T b$

$$\begin{pmatrix} 1-t \\ -1-t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}. \quad t \in \mathbb{R}.$$

$\Rightarrow \text{Null}(B^T B) = \text{Null}(B)$.

This gives all least squares sol. by thm.

Question 2 — solution: part 2

b_p = Au

By thm.

u is a least squares sol. $\iff$ Au is the orthogonal proj. of b onto $\mathrm{Col}(B)$.

So $b_p = Bu$ for any u we solved in Part 1

$$= \begin{pmatrix} -1 & 1 & 0 \\ 2 & 1 & 3 \\ -2 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix} \cdot \left(\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} -1 & 1 & 0 \\ 2 & 1 & 3 \\ -2 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ -4 \\ -3 \end{pmatrix} \neq \begin{pmatrix} -4 \\ 1 \\ -4 \\ -3 \end{pmatrix} = b.$$

$\uparrow$ $\mathrm{Null}(B)$

Conclusion:

- $b_p = \begin{pmatrix} -2 \\ 1 \\ -4 \\ -3 \end{pmatrix}$.
- Since $b_p \neq b$, $Bx = b$ is not consistent.

$b \notin \mathrm{Col}(B)$.

Section 5.5 Least Squares Solutions by QR Factorization

Theorem

Let A be an $m \times n$ matrix with linearly independent columns, with QR factorization $A = QR$, and let $\mathbf{b} \in \mathbb{R}^m$. Then $A\mathbf{x} = \mathbf{b}$ has a unique least squares solution,

$$\mathbf{u} = R^{-1}Q^T\mathbf{b}.$$

That is, it suffices to solve

$$R\mathbf{x} = Q^T\mathbf{b}.$$

pf

u is a l.s.s. to $Ax = b \iff A^TAu = A^Tb$ substitute $A = QR$

$\iff R^T(Q^TQ)Ru = (R^TQ^T)b$ ($Q^TQ = I$)

$\iff R^TRu = R^TQ^Tb$ (R is invertible $\Rightarrow$) $Ru = Q^Tb \iff u = R^{-1}Q^Tb$. □

Question 3

Let A be the matrix of Question 1. It has the QR factorization $A = QR$, where

$$Q = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ 0 & 2 & 2 \\ 2 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 3 & -3 & -3 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix}.$$

Let $\mathbf{b} = \begin{pmatrix} 3 \\ -1 \\ -5 \\ -1 \end{pmatrix}$. Use the QR factorization to find the least squares solution $\mathbf{u}$ of $A\mathbf{x} = \mathbf{b}$.

Question 3 — solution

Thm: u is "the" least squares sol. $\Leftrightarrow$ $u = R^{-1}Q^T b$

$Ru = Q^T b$. By Matlab, $Q^T b = \begin{pmatrix} -3 \\ -3 \\ 3 \end{pmatrix}$.

R | $Q^T b$

$$\left(\begin{array}{ccc|c} 3 & -3 & -3 & -3 \\ 0 & 3 & 3 & -3 \\ 0 & 0 & 3 & 3 \end{array}\right) \xrightarrow{\text{back substitution}} u = \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix}$$

$$\begin{cases} 3x - 3y - 3z = -3 \Rightarrow 3x = -3 + 3y + 3z = -3 - 6 + 3 = -6 \Rightarrow x = -2. \\ 3y + 3z = -3 \Rightarrow 3y = -3 - 3z = -6. \quad y = -2. \\ 3z = 3 \Rightarrow z = 1. \end{cases}$$

Next lesson

Next lesson (**Tue 20 Oct**, Section 6.1), I will focus on

1. the definition of eigenvalues and eigenvectors of a square matrix, and how to find them using the characteristic polynomial;

2. the **eigenspace** associated to an eigenvalue, and how to find a basis for it.

Please watch the video clips beforehand.